

Improved Batteries

In order to be completely free and move around as we please, we need to build better batteries. Decades ago, solar-power seemed to offer the promise of cheap and easily rechargeable batteries, but they just weren't efficient enough, and not to forget, useless at night. Currently, fuel cells are the most promising: they're cheap, use easily replenishable fuels, and already offer as much as 10 times more battery life and power than standard lithium-ion batteries.

Consider that we already have the capability of making laptops that are as powerful as the best desktops, and that mass usage of such devices is only stopped by the fact that we don't have decent batteries to support them. Future battery technologies will play a major role in making better and reliable mobile gadgets.

Software

Software, too, will have a huge role to play in our technological future. The idea of adaptive software is surfacing, and AI researchers are coming up with new ideas every day. If all goes well, you will soon be running your PC on software that can "think."

Wouldn't it be nice to wake up in the morning and find that your home's "central server" has already drawn your bath, chosen your favourite genre of news and made you the perfect cup of tea? Perhaps we will soon be able to do away with the keyboard and mouse and be able to converse with our PCs.

Research is underway to use a camera to follow eye movement and expressions, which will allow you to control a mouse cursor just by looking at it and mentally moving it to the desired place on the screen. Courtesy advanced algorithms, some robots can now learn, all by themselves, how to perform in the real world: they can evolve from scratch, growing up like a child.

Even weather forecasting systems will finally become reliable, crunching gargantuan amounts of data to predict precisely how many inches it will rain in a given hour. Number crunchers will also completely map the human genome, and there's no limit to what researchers can do given that mapping. Similarly, there's no



Several years ago, even X-Rays were considered to be high-tech. Today, there are telesurgeries conducted with the help of robots, so, a doctor's physical presence might not be required in the near future

telling what will happen when software is able to predict exactly how much a certain stock will be worth on a certain day!

Medical Technologies

In order to improve our lives, we need to first do away with suffering and premature death. *Digit* has talked about cyborgs earlier, and advancements in cybernetics offer a lot of hope. Most of us would readily trade in all material comforts to be able to live 10 or 20 years longer, and cybernetics makes this a very real possibility.

Imagine a person suffering from lung cancer; the standard procedure is to undergo chemotherapy, which often does more harm than good. If cybernetics is perfected, all the patient would need to do is undergo lung transplant surgery and get fitted with synthetic lungs. Arteries, veins and even muscles could just be rebuilt using plastics.

There's also the developments in areas such as telesurgery, where the surgeon sits at a workstation with mechanical arms, and has a 3D view of the surgical site. Telesurgery could be performed on patients at the other end of the world. Then there's cryonics—the science (or science fiction) of keeping a person alive in a


➔


Check-in at www.jetairways.com
Print your own Boarding Pass.



- You can check in from wherever, at your convenience • Select your own seat
- Print your own Boarding Pass • Walk straight to security check with hand baggage



the Spirit of New India

Web Check-in opens 48 hours prior to departure. For more details, login to www.jetairways.com Book your e-Seat at Jet Airways city offices/airport counters, through your nearest travel agent or at www.jetairways.com.

NETWORK ● JAN 06-05